GGG-C-743C
March 31, 1971
SUPERSEDING
Fed. Spec. GGG-C-743B
March 6, 1968

FEDERAL SPECIFICATION

CUTTERS, BUTTER, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of hand-operated butter cutter.

1.2 Classification. Butter cutters shall be of the following sizes as specified (see 6.2):

Size 60 - 60 pats per pound.
Size 72 - 72 pats per pound.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-A-591	-	Aluminum Alloy Die Castings.
QQ-A-601	-	Aluminum Alloy Sand Castings.
QQ-W-423	-	Wire, Steel, Corrosion-Resisting.
UU-P-268	-	Paper, Kraft, Untreated, Wrapping.
PPP-B-636	-	Box, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washing, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Sanitation Foundation (NSF):

Basic Criteria No. C-2 - Special Equipment and/or Devices.

Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, P. O. Box 1468, Ann Arbor, Michigan 48106.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The butter cutters delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with specification requirements.

3.2 Standards compliance. The butter cutters shall conform to the applicable requirements of NSF Basic Criteria No. C-2.

3.2.1 NSF certification. The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the butter cutters he proposes to furnish under this specification meets the applicable requirements of NSF Basic Criteria No. C-2 as follows:

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(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF Seal on the finished butter cutter shall be submitted prior to approval of the first shipment, or

(2) A certification for the butter cutter issued by NSF under their special one-time contract evaluation/certification service shall be submitted prior to approval of the first shipment, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical authority functioning in advisory capacity to the preparing agency of the specification indicating that the butter cutters have been tested and conform to NSF Basic Criteria No. C-2 shall be submitted prior to the approval of commencing production.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his butter cutters, provided the completed items comply with all provisions of this specification. Except as specified herein, commercial tolerances are applicable to all material thicknesses and diameters.

3.3.1 Aluminum. Aluminum shall conform to alloy 214, temper F, of QQ-A-601 or alloy 380 of QQ-A-591.

3.3.2 Corrosion-resisting steel wire. Corrosion-resisting steel wire shall conform to form I, composition 302 or 304 of QQ-W-423 except the minimum tensile strength shall be 304,000 pounds per square inch (p.s.i.). The diameter of the wire shall be not less than 0.018 nor more than 0.020 inch.

3.4 Design and construction. The butter cutters shall be designed as specified herein and as illustrated on figure 1. (Figure 1 is used for illustration purposes only and is not intended to preclude the purchase of butter cutters which are otherwise in accordance with this specification.) The size 60 and 72 butter cutters shall be capable of cutting a 1-pound block of butter measuring 2-1/2 inches square and 4-3/4 inches in length into 60 and 72 equal size pats, respectively, in two operations. The butter shall be cut into the number of specified pats by one horizontal motion of the cross-cutter and one downward motion of the top-cutter and the pats shall be smooth and uniform when tested in accordance with 4.4.1. The butter cutter shall consist of a base and vertical post upon which are mounted a butter holder, a top-cutting frame, and a cross-cutting frame. The frames shall be provided with cutting wires for producing the required number of pats.

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3.4.1 Base. The base shall be made from iron or steel with a carbon steel vertical post having a minimum diameter of 1/2 inch and a minimum length of 8 inches securely bolted on the underside at one end of the base. The post shall be at a right-angle to the base and may be shouldered-down to permit proper seating in the base.

3.4.2 Butter holder. The butter holder shall be made from aluminum specified in 3.3.1, bronze, or die-cast zinc, and shall consist of a horizontal rectangular plate and a vertical square plate, sized and joined to accurately fit one side and end of the 1-pound block of butter specified in 3.4. The butter holder shall be integrally attached to the lower end of the post, so that its long axis corresponds to that of the base. The flat surfaces of the butter holder coming in contact with the butter shall be corrugated or roughened to prevent butter from slipping and shall be of a design to permit ease in cleaning. The horizontal rectangular plate shall be provided with a straight central longitudinal slit for passage of the cross-cutting frame vertical wire, and the long edge of the plate shall have a flange or track for passage of the sliding cross-cutting frame.

3.4.3 Cross-cutting frame. The cross-cutting frame shall consist of a square-shaped frame of aluminum specified in 3.3.1, bronze, or die-cast zinc, sized to prevent contact with the surfaces of the butter. The bottom of the cross-cutting frame vertical sides shall be furnished with a lip or guide that will allow the cross-cutting frame to slide along the tracks or flanges of the horizontal rectangular plate without binding. A vertical cutting wire passing through the slit of the plate, and a horizontal cutting wire shall be tightly stretched across the center of the frame in a manner to insure accurate cutting of a block of butter into four equal bars when the frame is pushed longitudinally.

3.4.4 Top-cutting frame. The top-cutting frame shall be a rectangular-shaped frame of aluminum specified in 3.3.1, bronze, or die-cast zinc. The frame shall be provided at one end with a sleeve to fit the vertical post and slide freely upon it. The cutting wires shall be tightly stretched across the frame at equal intervals so as to cut the block of butter vertically into the desired number of equal size pats. The tightly stretched cutting wires shall be secured to the frame by setscrews, adjusting screws or tightening screws, and bushings. The screws shall not be used in the food product zone.

3.4.5 Wires. The cutting wires of the top-cutting frame and cross-cutting frame shall be wire specified in 3.3.2, and shall be removable for replacement in frames.

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3.4.6 Metal strips. Metal strips, when used to cover bushings or set-screws for wire tension, shall be fabricated of corrosion-resisting steel, carbon-steel, brass, or bronze.

3.4.7 Screws. All screws shall be fabricated of corrosion-resisting steel, carbon-steel, brass, or bronze.

3.4.8 Bushings. Bushings shall be fabricated of corrosion-resisting steel, brass, or bronze, and shall not extend above the surface of the frame after the wires are tightened.

3.5 Marking for identification. The butter cutter shall be permanently marked with the manufacturer's name, tradename, or trademark of such known character as to be readily identifiable with said manufacturer.

3.6 Finish. With the exception of the base, all parts not fabricated of corrosion-resisting steel or aluminum shall have a polished chromium or nickel-plated finish. The chrome or nickel-plating shall not show any signs of pitting, blistering, peeling, or eruption. All aluminum parts shall be anodized in accordance with type II, class 1 of MIL-A-8625. The base shall be finished with baked enamel in accordance with the manufacturer's standard practice. The baked enamel finish shall not be chipped, blistered, peeled, or have any area of no finish. Cadmium platings and baked enamel finish shall not be used in the food product zone.

3.7 Workmanship. The finished butter cutter shall be clean, and shall not contain any rough or sharp edges, burrs, slivers, rough die or grind marks, gouges, rust, or scale. Castings shall not be cracked, porous, warped, or show any signs of miscasting or blowholes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Inspection. Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall consist of all butter cutters of one size offered for inspection at one time. The sample unit shall be one completely fabricated butter cutter.

4.2.2.1 Visual examination. Examination shall be made of the butter cutter for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish			
Chromium or nickel-plated components	Not a polished chrome or nickel-plated commercial finish (where required)		X
	Plating peeled, blistered, erupted, or pitted	X	
Enamel components	Any evidence of chipping, blistering, peeling, or area of no film	X	
Construction and workmanship	Any component not securely attached (i.e. non-movable parts)		X
	Functioning component inoperative (i.e. horizontal or vertical frames)	X	
	Any functioning component that required excessive pressure to operate		X
	Any rough or sharp edges, burrs, slivers, rough die or grind marks, gouges, rust, or scale	X	

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TABLE I. Classification of defects (con.)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (con.)	Casting cracked, porous, or warped	X	
	Casting miscast or contains blowholes	X	
Base	Post not at right angle to base	X	
	Base missing	X	
	Post missing	X	
Butter holder	Horizontal, rectangular, or vertical square plate missing	X	
	Butter holder not integrally attached to post	X	
	Flat surfaces not corrugated or roughened (i.e. where surface contacts butter)	X	
	No straight longitudinal slit on horizontal rectangular plate	X	
	Flange or track not provided for cross-cutting frame	X	
Cross-cutting frame	Cross-cutting frame missing	X	
	Cross-cutting frame sides not provided with lips or guides to slide in the tracks or flanges of horizontal rectangular plate		X
	Cross-cutting frame does not cut butter into four equal bars		X
Top-cutting frame	Frame missing	X	
	Not provided with hole for post	X	
	Cutting wires not equally spaced	X	
	Setscrews or bushings protrude above frame surface		X
Threaded fasteners	Missing, broken, stripped, or fractured	X	
Identification marking	Missing, illegible, incomplete		X

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4.2.2.2 Dimensional examination. Examination shall be made of the butter cutter for compliance with dimensions specified herein. Any nonconformance shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. The initial production unit shall be tested as specified in 4.4.1. Any nonconformance shall be cause for rejection of the lot.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Markings	Missing, incorrect, illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.
Contents	Number per container is more or less than required.

4.2.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.2.3.2 Examination of palletization. An examination shall be made to determine that palletization is in compliance with Section 5 of this specification. Defects shall be as indicated in table III. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

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TABLE III. Examination of palletization

Examination	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.3 Standard compliance. Proof of compliance with the requirements of 3.2 shall be submitted to the Government representative.

4.4 Tests.

4.4.1 Operational test. The butter cutter shall be loaded with a pound of butter measuring 2-1/2 inches square by 4-3/4 inches in length and having a temperature between 40°F and 50°F. The two cutting operations shall be performed, and the butter pats shall be checked to determine compliance with 3.4. Inability to meet the requirements specified shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each butter cutter shall be completely assembled, and then be completely wrapped in minimum 30-pound kraft paper conforming to grade B of UU-P-268, and then packaged upright in a snug-fitting fiberboard box conforming to style RSC, type CF, variety optional, or type SF, class domestic, grade 200 of PPP-B-636. The butter cutter shall be held securely in place within the box by means of scored and folded pads or die-cut sheets made of the same material as the box. The design, number, and placement of the inner packing shall eliminate movement of the butter cutter or parts, provide clearance for projecting parts from the inner walls of the box, and prevent damage to the parts from shock or vibration while in transit. Each box shall be securely closed in accordance with the appendix of the box specification (see 6.3).

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5.1.2 Level C. Butter cutters complete, shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve butter cutters of one size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636, except that the weight limitations shall not apply. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Civil agencies. Four butter cutters of one size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method I of the appendix of PPP-B-636.

5.2.2.2 Military agencies. Twelve butter cutters, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety optional, or type SF, class domestic of PPP-B-636, except that the weight limitations shall not apply. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.2.1 When specified (see 6.2), the shipping container shall be a V3c, V3s or V4s fiberboard box fabricated and closed in accordance with PPP-B-636.

5.2.3 Level C. Butter cutters complete, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization (Military requirements). Unless otherwise specified (see 6.2), butter cutters packed as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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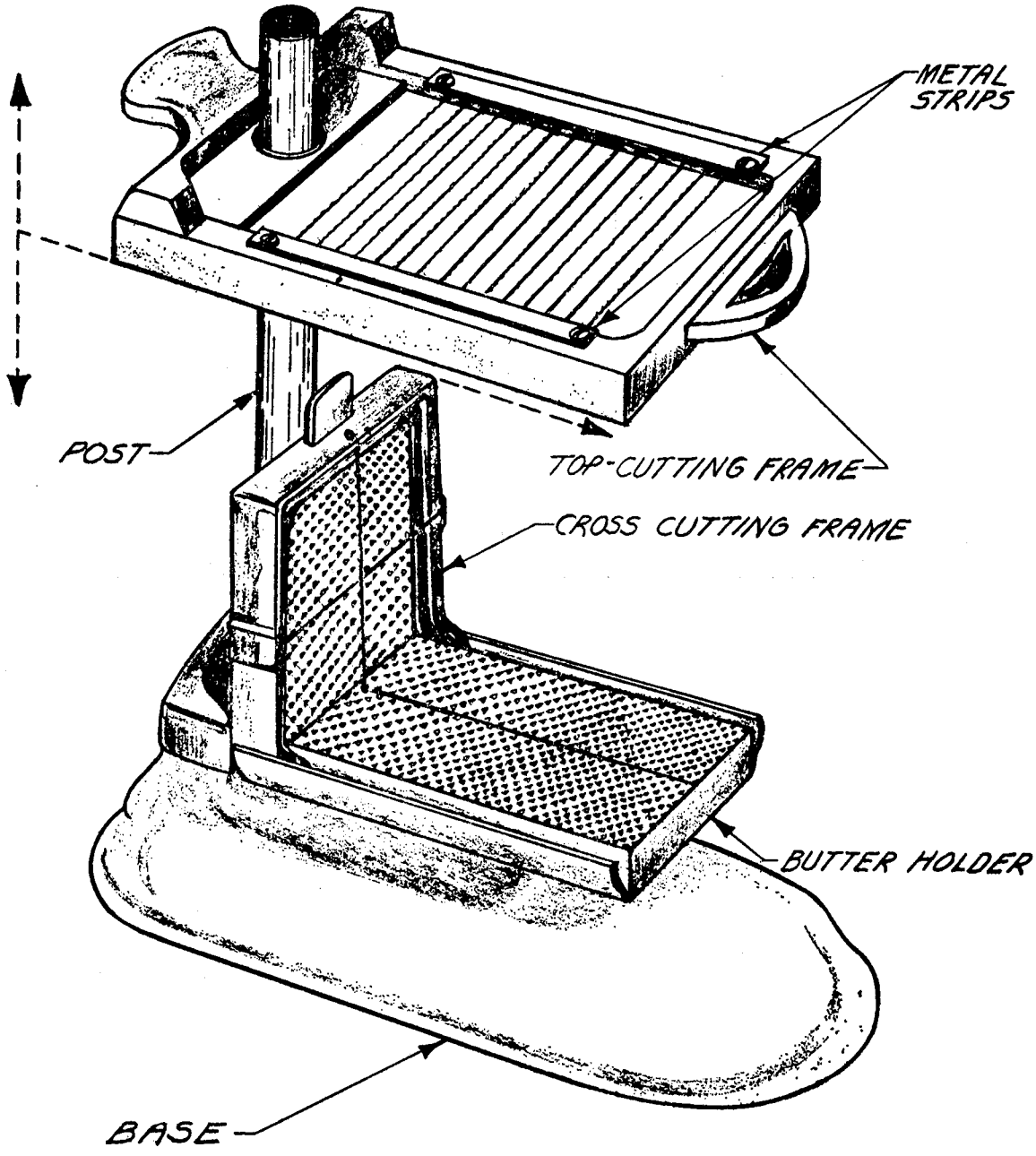


FIGURE 1.—Cutter, butter, hand.

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5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The butter cutters covered by this specification are intended for use in military and institutional kitchens to cut a pound of butter into equal pats for serving.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of butter cutter required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Whether weather-resistant grade fiberboard shipping containers are required for level B Military packing (see 5.2.2.2.1).
(e) When palletization is not required (see 5.3).

6.3 Packaging. The method of packaging referred to in 5.1.1, is similar to method III of MIL-P-116.

6.4 For civil agency procurements. When level B packing is required for civil agency procurements, the level A requirements in 5.1.1 shall apply.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

Army - GL

Project No. 7330-0287

Review activities:

Army - MD
Navy - MS

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